

Writing a good protocol

The purpose of a protocol is simple: to make experiments easier to perform, easier to share, and easier to build on. However, a good protocol is more than just a record of how something is done. Good protocols directly improve reproducibility, efficiency, and collaboration. They also reduce mental clutter and make space for creative thinking (Anderson, 1982; Sweller et al., 1998).

Here you'll find practical tips for writing protocols that age well. The advice is format-agnostic: whether you write in Markdown, structured XML, or word-processor documents, the same discipline applies.

Procedures

>> Writing a protocol

- (1.) Sketch the structure first. At minimum: a unique identifier, a name, an abstract, a changelog, and one or more procedures. Add safety, analysis, or troubleshooting advice where applicable.

This is why: A consistent skeleton across your collection lets readers find “how do I troubleshoot this?” or “what changed in the latest version?” without rereading the whole document.

- (2.) Write each step as a single action in active voice.

- (3.) Be deliberate:

- Name equipment, reagents, and durations a new reader would need to execute the step without asking you. Assume nothing about prior knowledge — including your own, two years from now.
- Parameterize quantities, volumes, and ratios so the procedure scales. A protocol is not a lab notebook. Leave room for the next user to adjust without rewriting it.
- When an action is non-obvious or grounded in experience, attach a one-sentence rationale.

This is why: Stating the “why” raises adherence (Deci et al., 1994) and lets the next user adapt the step to new conditions instead of preserving it as cargo cult.

- (4.) *Critical:* Mark critical steps and decision points so they are visually distinct from routine ones. Readers scan; the moments that matter most should be impossible to skim past. 

- (5.) Track changes:

- Maintain a changelog. Append one line per substantive edit with the date, your full name, and a one-sentence summary of what changed.

Hint: Use full names (no initials). Minor corrections (patches) may not need to be logged. Backward-compatible additions and clarifications get a log entry. Incompatible changes get a new version, not an in-place edit.

- Version the filename. For incompatible changes, create a new file with an incremented version number and leave the retired version intact for reproducibility.
- *Optional:* Commit changes through version control  SOP0101. 

This is why: Version control records who reviewed what, when, and why. Such a paper trail is often essential when reproducing or auditing past work.

- (6.) Give credit:

- Audit AI-assisted content. If a language model drafted or revised any section, verify every factual claim and citation before publishing. Remove anything that can't be confirmed.
- Do not copy-paste copyright-protected material without permission or a compatible license. If you adapt content from another protocol, cite it and confirm its license is compatible with the one you intend to publish under.

- (7.) Review protocols regularly with users and collaborators. A second reader catches assumptions the writer is blind to.

List of references

- J.R. Anderson, *Psychol. Rev.* **89**(4), 369–406 (1982). J. Sweller, J.J.G. van Merriënboer, and F.G.W.C. Paas, *Educ. Psychol. Rev.* **10**(3), 251–296 (1998).
E. Deci, H. Eghrari, B. Patrick, and D. Leone, *J. Pers.* **62**(1), 119–142 (1994).

Change log

- | | | |
|------------|------------------------|--|
| 2025-04-02 | Benjamin C. Buchmüller | Initial version. |
| 2025-08-14 | Benjamin C. Buchmüller | Added sections on XML validation and organization using Git or GitHub; ChatGPT 5 (OpenAI) advised on XSD and Schematron use (XSLT-1 skeleton). |
| 2026-05-12 | Benjamin C. Buchmüller | Promoted to public; scope reduced to format-agnostic guidance. XML/ConTeXt-specific authoring content moved to an internal reference document. Guidelines block folded into the procedure as concrete writing actions. |

Open Protocol — Part of the *Lab Protocols* collection (2025) by B. C. Buchmüller and contributors. This document is made available under the Creative Commons Attribution Share Alike 4.0 International License. To view a copy of this license, visit <https://creativecommons.org/licenses/by-sa/4.0/>.

For research use. Provided in good faith, without warranty or liability for any use or results. Users are responsible for compliance with local regulations and institutional policies.

Current when printed. Visit <https://benjbuch.github.io/check/> or scan the QR code to check for updates.



17ad3a0